

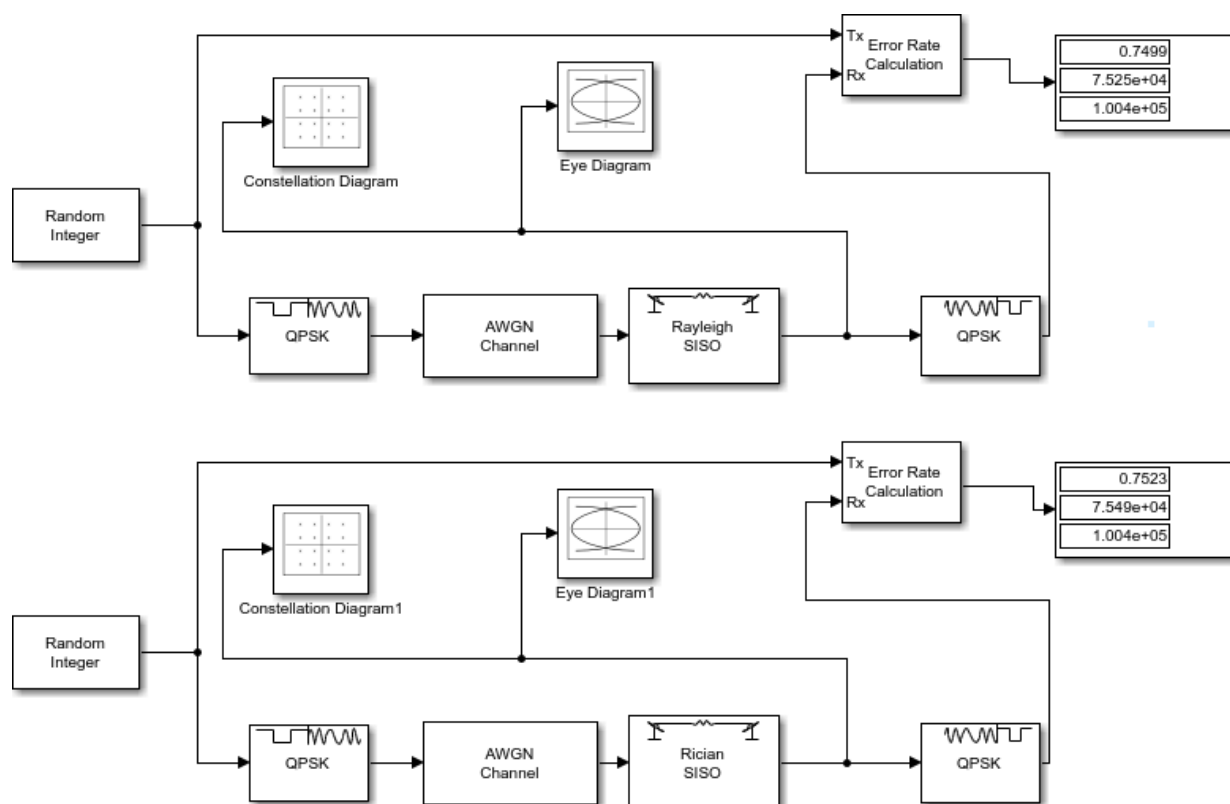
Lab 10: Rayleigh and Rician Fading

Objective: The aim of this Lab is to simulate and analyze the impact of different channel models on a communication system.

1. Channel Modeling:

Fading Channels:

- Simulate fading channels such as Rayleigh or Rician fading to represent multipath propagation and signal fading.



- Observe the variations in received signal strength over time.

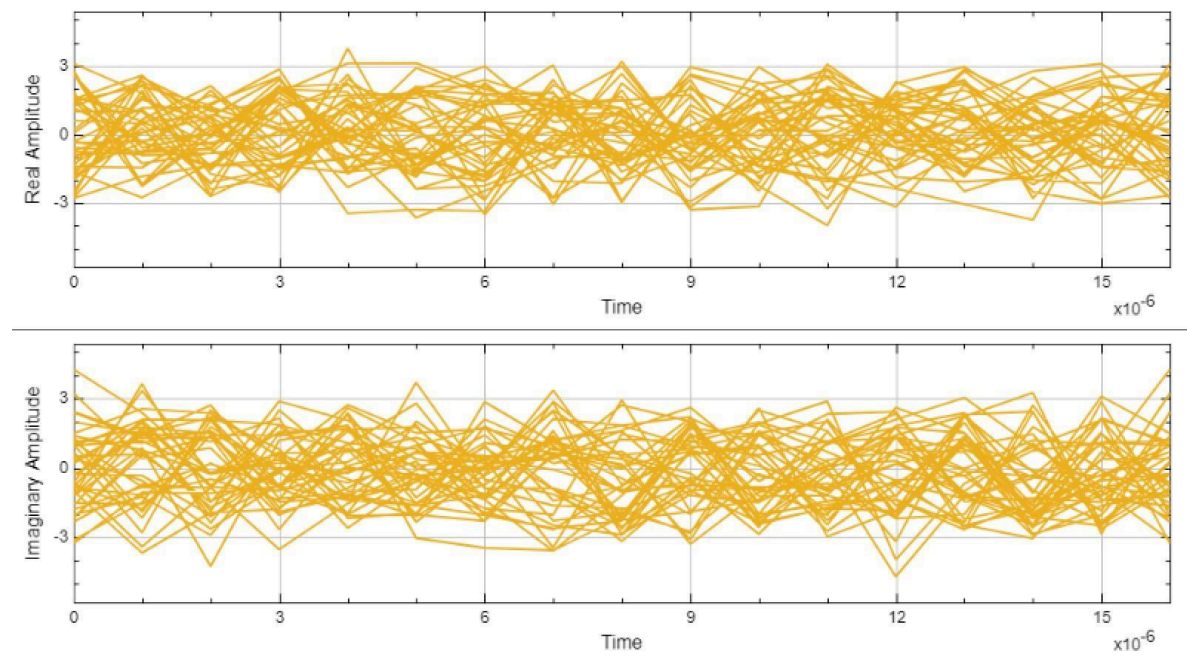
2. System Performance Evaluation:

a. Eye Diagrams:

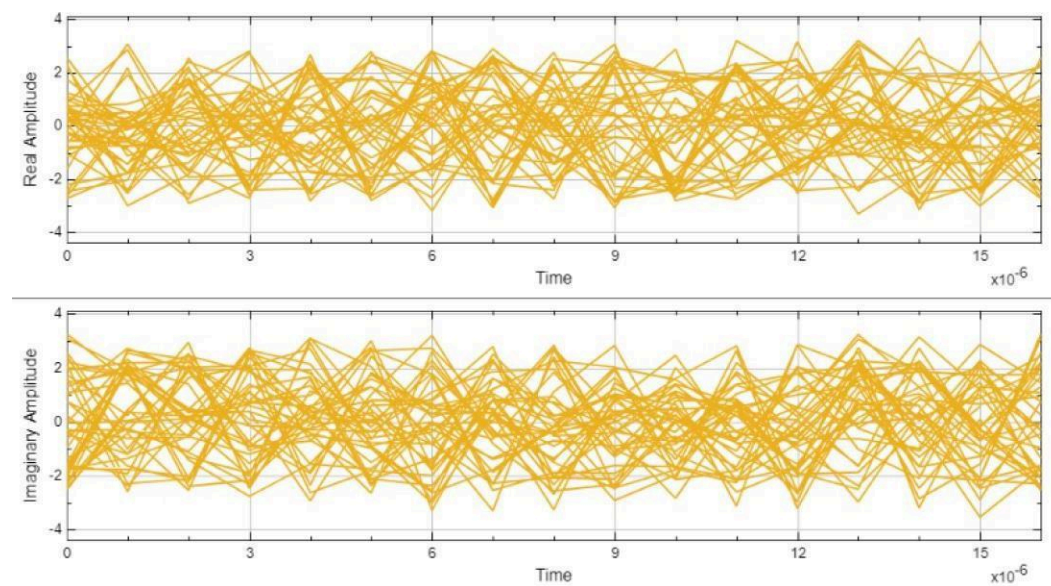
- Generate eye diagrams to visualize the quality of the received signal.

Rayleigh: Here, the absence of the LOS signal makes the zigzag design hardly noticeable. Eye closing vertically indicates that our signal is now extremely noisy, while narrow horizontal opening indicates severe inter-symbol interference (ISI).

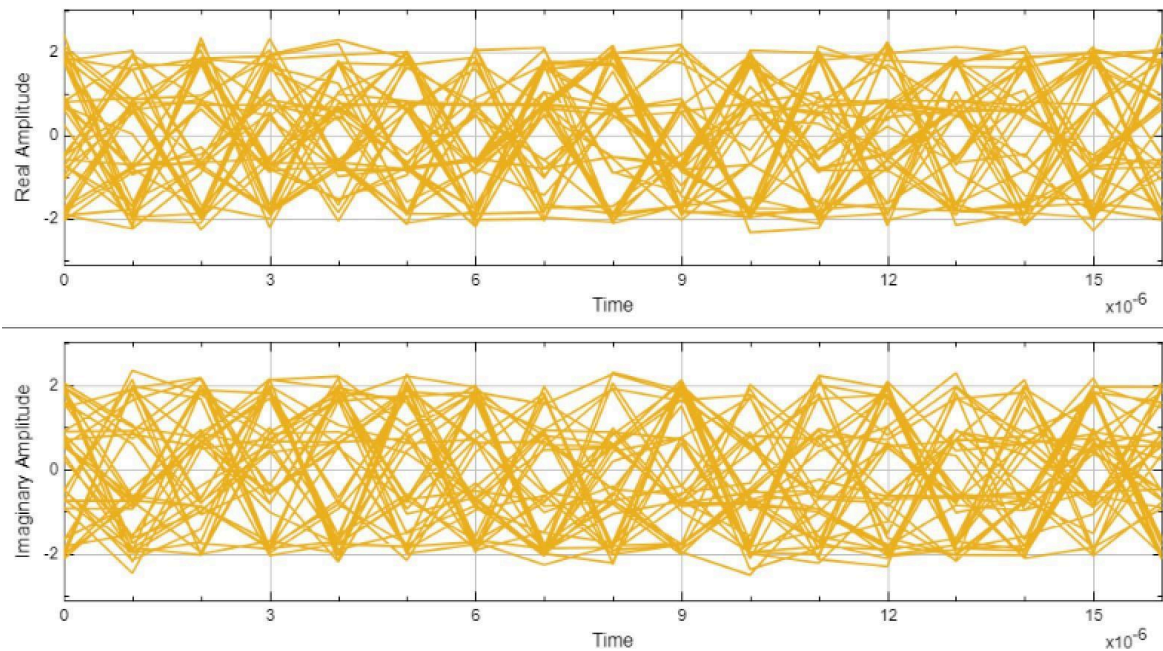
For $E_b/N_0 = 3$:



For $E_b/N_0 = 10$:

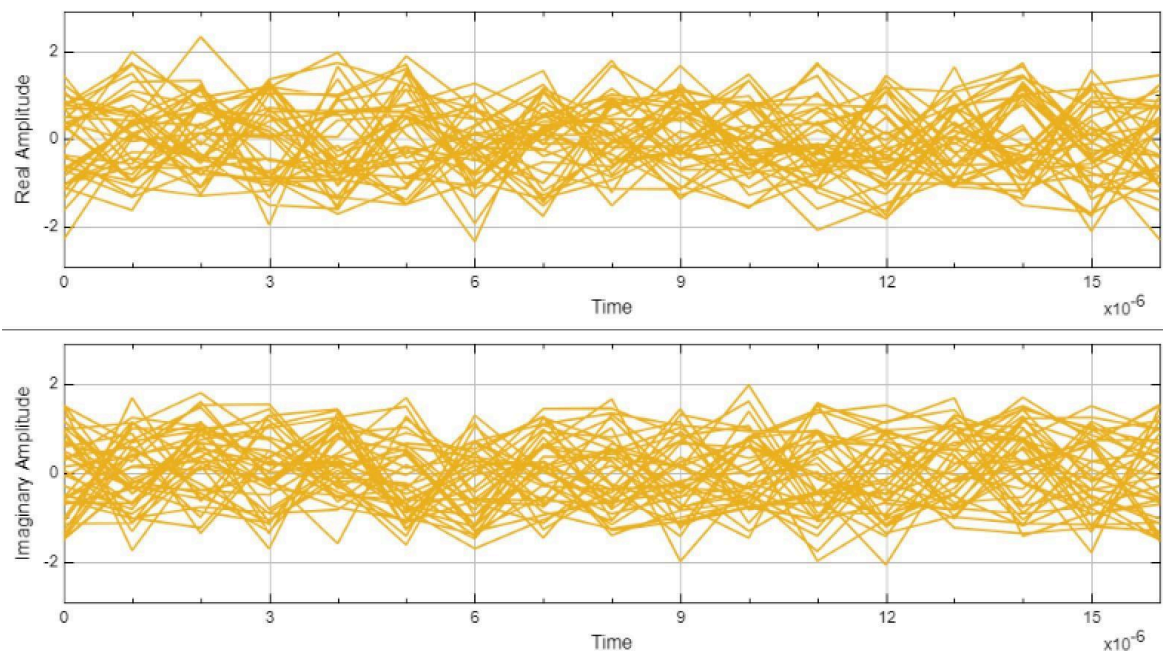


For $E_b/N_0 = 15$:

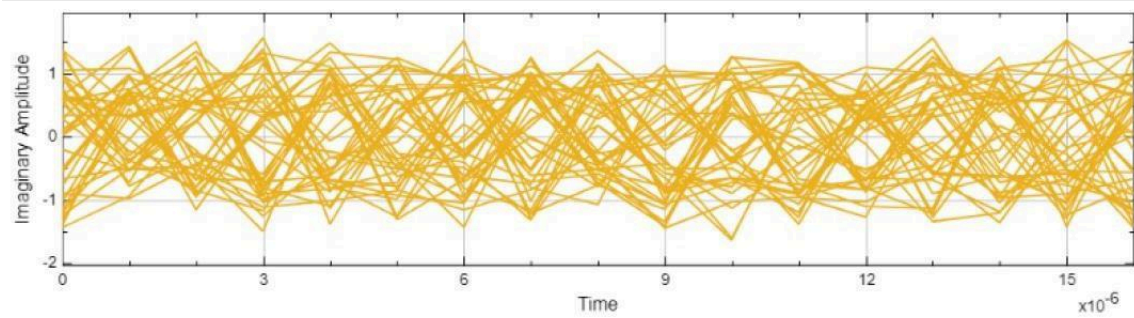
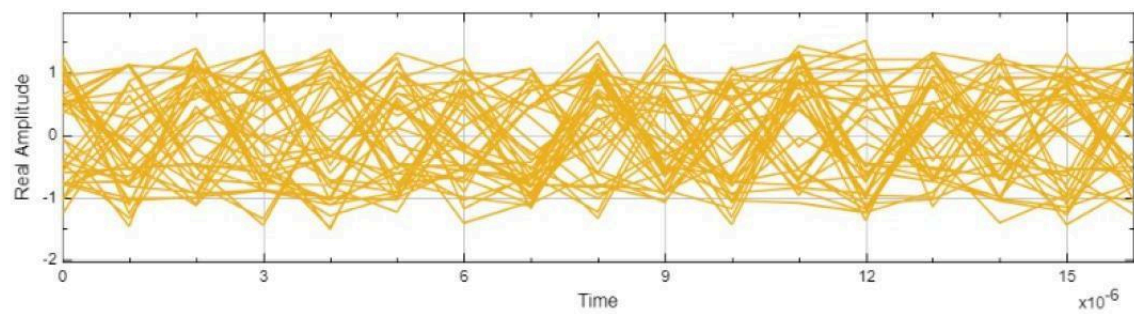


Rician: Although the eye diagrams are now considerably more messy, a zigzag diagram is still visible. increasing ISI has caused the breadth to decrease, while increasing noise has caused the height to decrease. even if the LOS path dominates and we can still see the general outline.

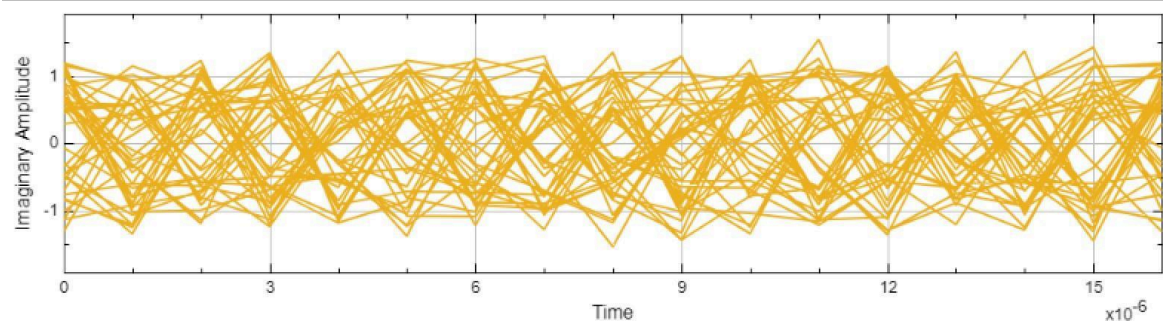
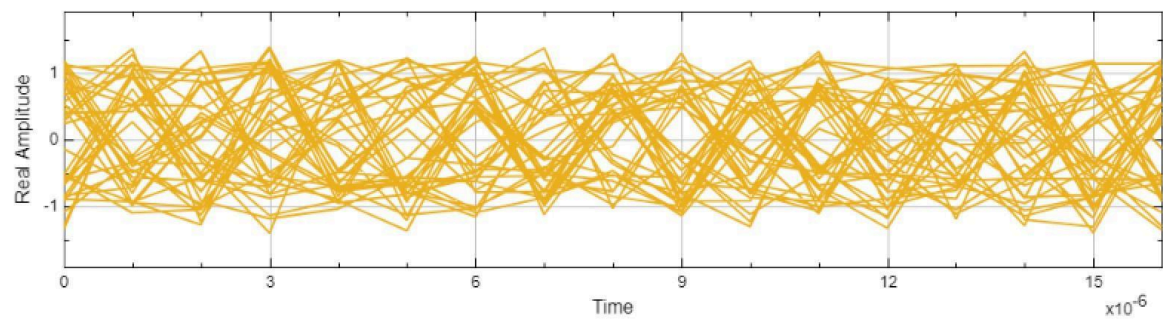
For $E_b/N_0 = 3$:



For $E_b/N_0 = 10$:

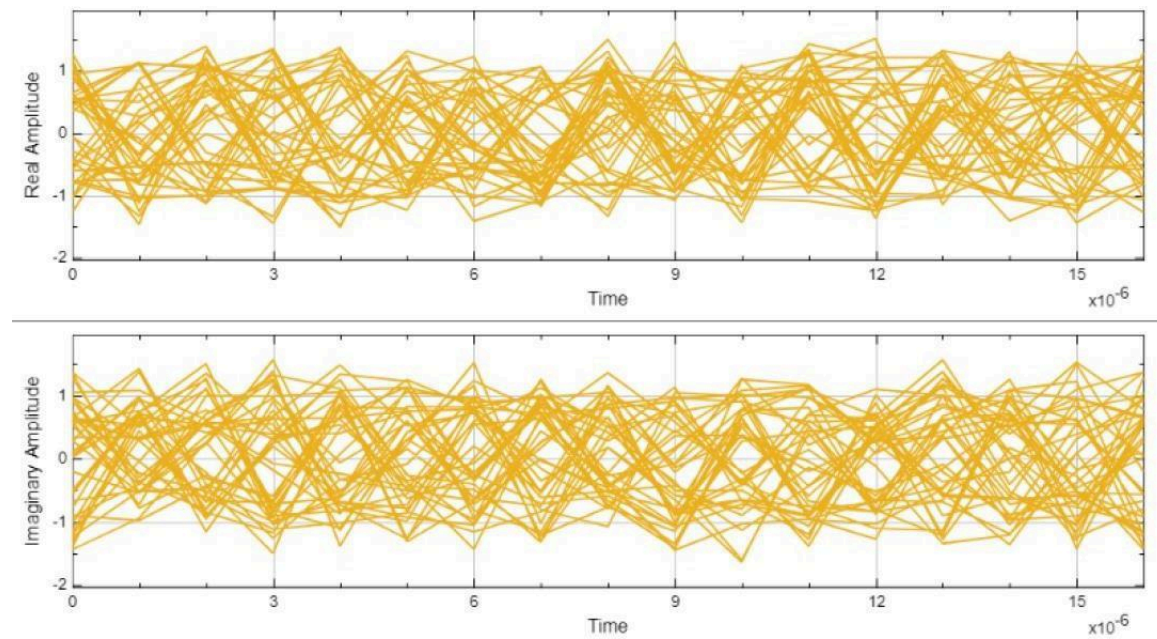


For $E_b/N_0 = 15$:

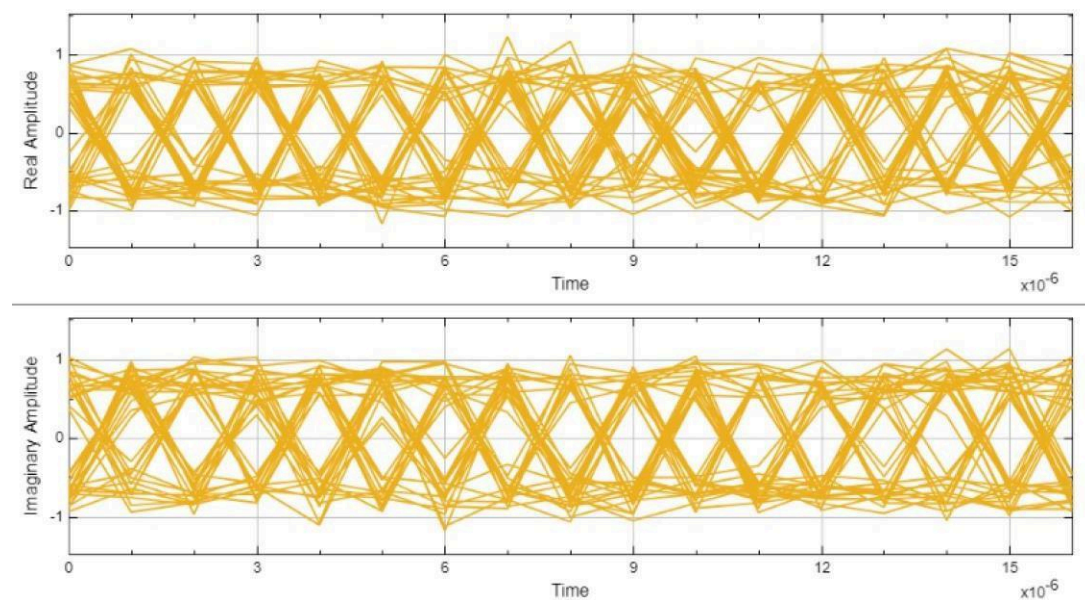


No Fading: Very clear eye diagram, denoting no effect of fading.

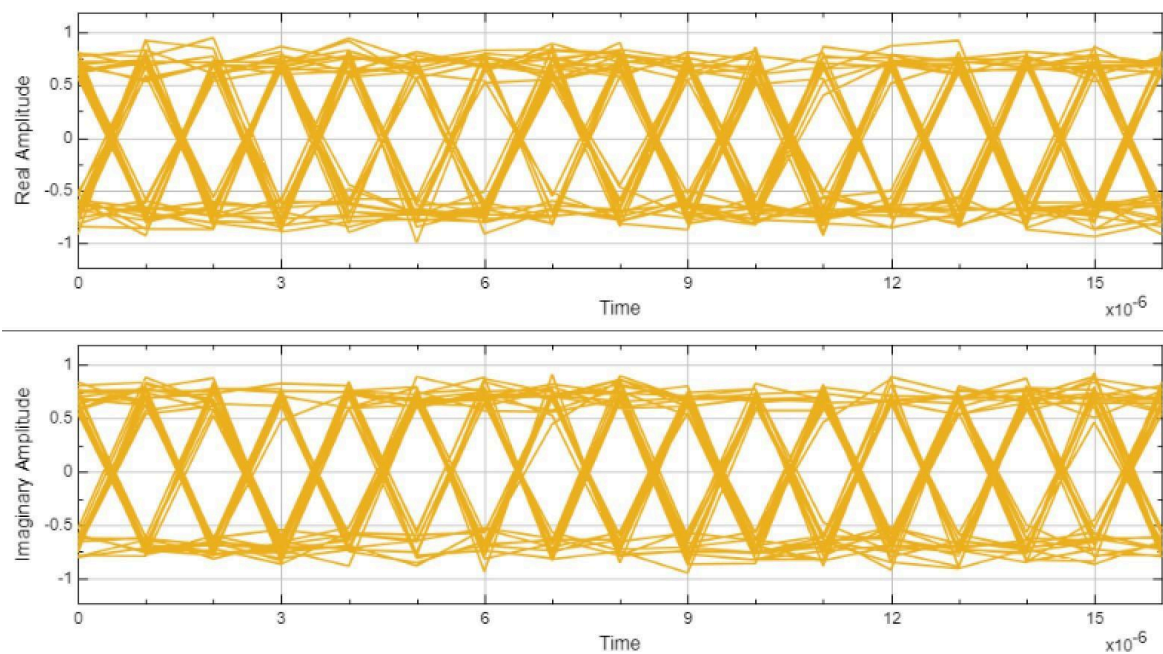
For $E_b/N_0 = 3$:



For $E_b/N_0 = 10$:



For $E_b/N_0 = 15$:

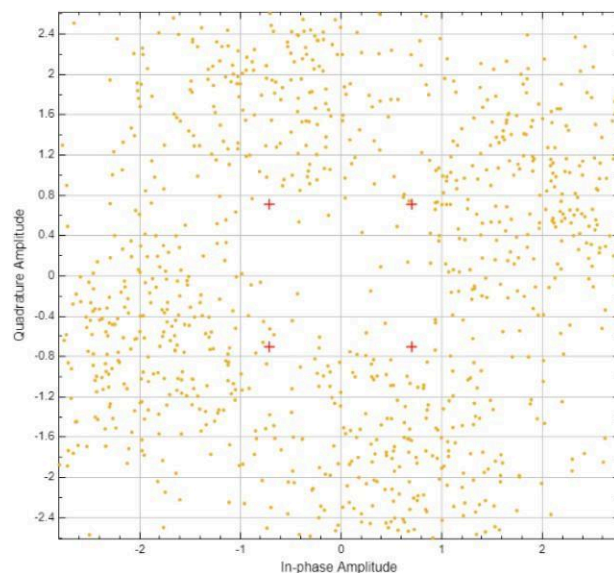


b. Constellation Diagrams:

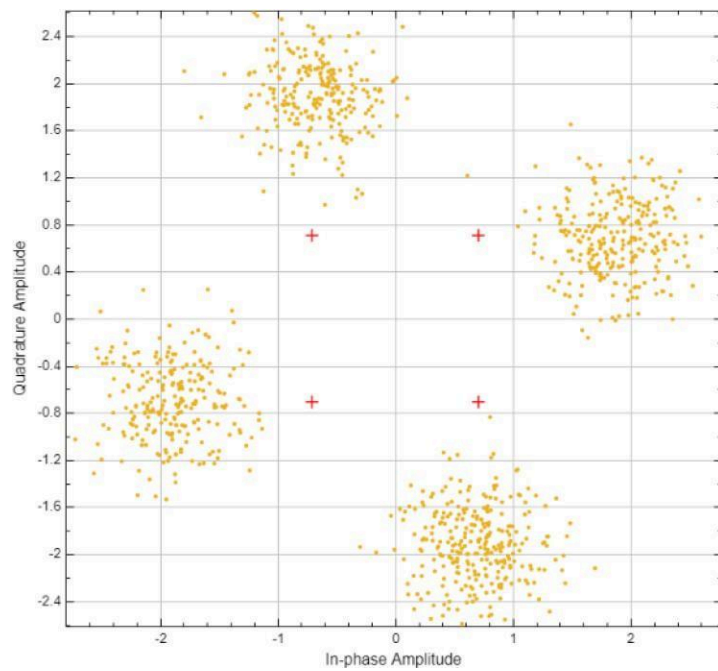
- Plot constellation diagrams to analyze the distribution of points in the signal space.

Rayleigh: The scattered points that we saw in Rician fading now have incorrect phases due to no LOS signal. So, they are rotated by a phase shift.

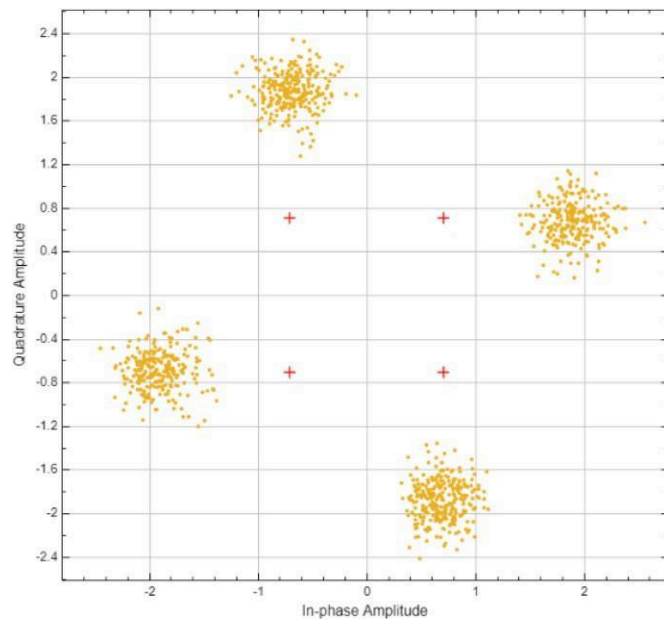
For $E_b/N_0 = 3$:



For $E_b/N_0 = 10$:

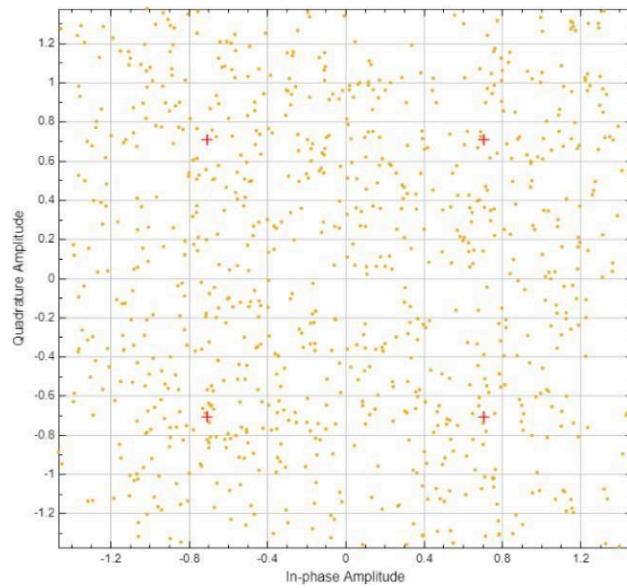


For $E_b/N_0 = 15$:

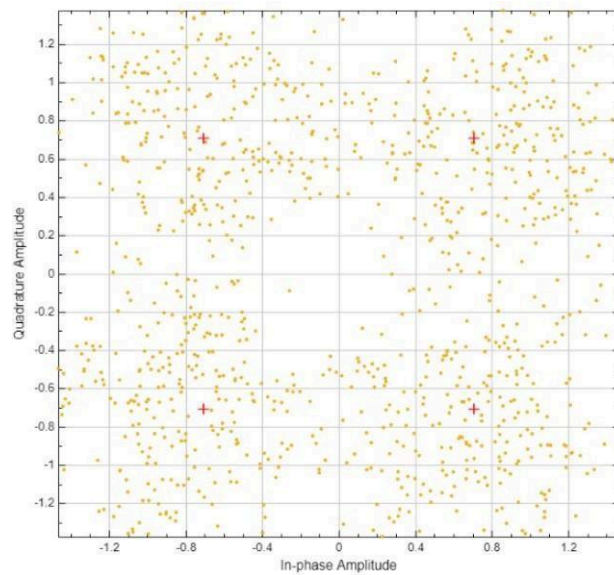


Rician: Much greater scattering due to interference from multipath fading, but still barely around ideal points.

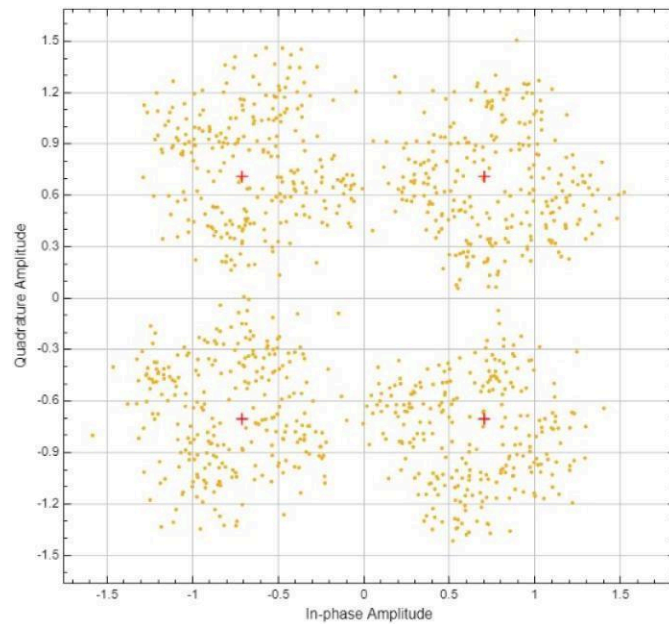
For $E_b/N_0 = 3$:



For $E_b/N_0 = 10$:

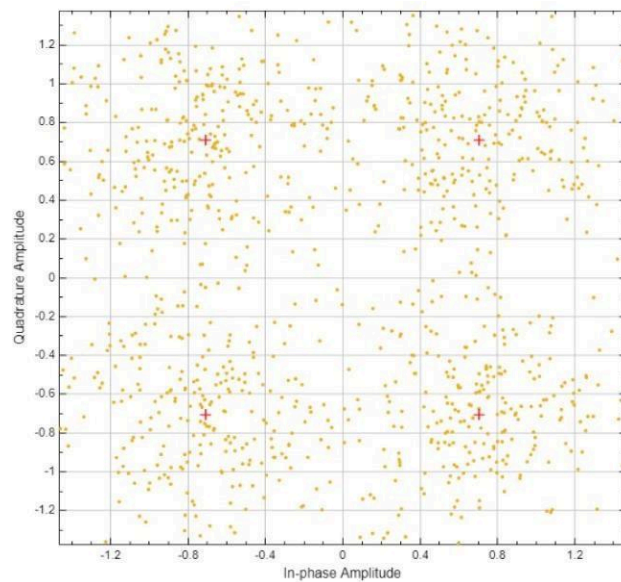


For $E_b/N_0 = 15$:

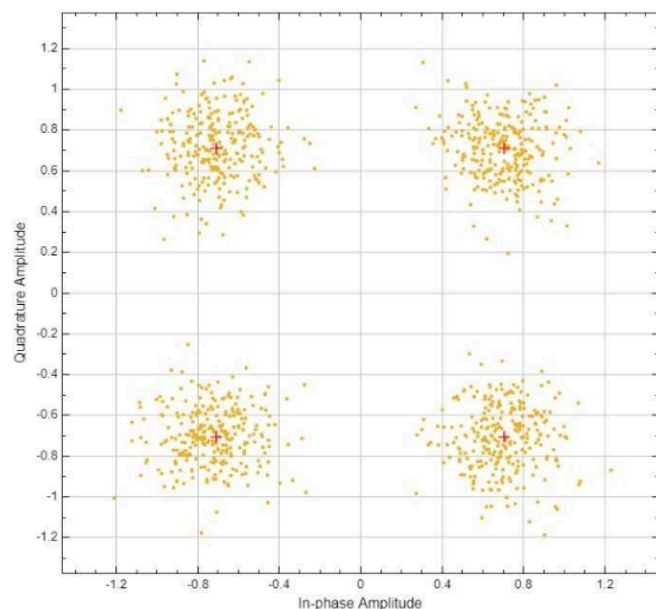


No Fading: Tight clustering around the ideal positions, denoting stability (low amplitude and phase variations)

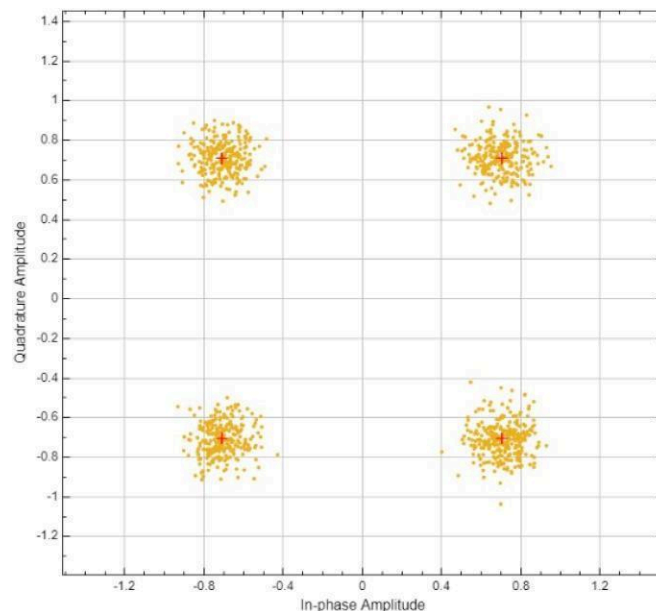
For $E_b/N_0 = 3$:



For $E_b/N_0 = 10$:



For $E_b/N_0 = 15$:



c. Bit Error Rate (BER) Analysis:

- Measure the BER of the communication system under different channel models

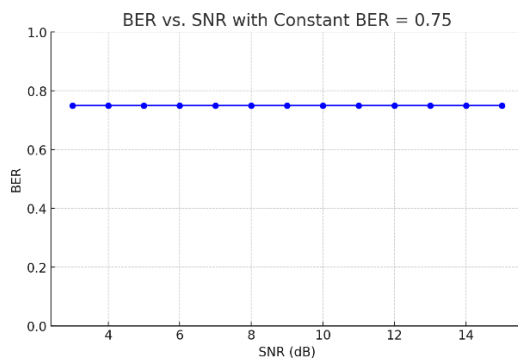
Because we are unable to properly demodulate it because of multipath fading and the lack of equalisation procedures, the BER is always high. Therefore, we obtain bad BER for both situations despite the differences in the diagrams; once we attempt equalisation, we will determine the difference between the two.

Eb/No	BPSK BER	QPSK BER
3	0.75	0.75
4	0.75	0.75
5	0.75	0.75
6	0.75	0.75
7	0.75	0.75
8	0.75	0.75
9	0.75	0.75
10	0.75	0.75
11	0.75	0.75
12	0.75	0.75
13	0.75	0.75
14	0.75	0.75
15	0.75	0.75

d. SNR vs. BER Performance Curves:

- Construct performance curves and tables that illustrate the relationship between SNR and BER for different channel conditions.

For both channels the following is observed.



3. Documentation and Analysis:

a. Report Writing:

- Document the labelled Simulink Model including parameters setting values.
- Provide an analysis of the results, highlighting the performance of the communication system under various conditions.
- Attach all the Diagrams, tables and Graphs in Report.

- Parameters followed:

Rayleigh: Doppler Shift: A change in frequency brought on by relative motion that influences the pace at which signals fluctuate. 200 Path Delays: Shows each reflected path's time delay. $\ast[0\ 0.2\ 0.4\ 0.8] = 1.0\text{e-}6$
Average Path Gains: Signal power levels are influenced by the relative power of each path. [0–3–6–9]

Rician: K-Factor: Stronger LOS presence is indicated by higher values. 10 Doppler Shift, Path Delays, and Average Path Gains: Similar to Rayleigh, but taking the LOS component into consideration. Doppler shift of LOS = 10.

- Analysis:

Despite the QPSK modulation, the communication channel becomes quite poor under Rayleigh fading. This is because there are multiple routes—none of which are LOS—by which our signal is getting to the target. As a result, it bounces off objects like trees and buildings and meets at the end, which causes issues with its phase, amplitude, and synchronisation, to mention a few.

Since Rician fading now includes the straight LOS signal, which is what we really want to receive exclusively, it performs better in the eye and constellation diagrams. Nevertheless, the output is still sufficiently distorted by the addition of the NLOS signals for the QPSK system to function as poorly as during Rayleigh fading. After applying equalisation procedures to both fading techniques, I believe we will observe a difference in BER outputs, with Rician fading performing significantly better than Rayleigh fading.

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Date: 19-11-25

EE-322L Analog and Digital Communications Lab
Fall 2025
Habib University
Dhanani School of Science & Engineering



LAB 10 Rayleigh and Rician Fading

Lab #10 Marks distribution:

		LR1=20	LR3=10	LR4=30	LR5=30	AR4=20
In-Lab Tasks	Task 1	/10	/05	/10	/15	/20
	Task 2	/10	/05	/10	/15	
Marks Obt.	/100					

Lab Evaluation Assessment Rubric

#	Assessment Elements	(0<Level 1<=4)	(4< Level 2<=6)	(6< Level 3<=8)	(8< Level 4<=10)
LR1	Neat and Clean Circuit layout	Components are wired and didn't show neat and clean connections and minimal efforts shown.	Components are wired with untidy connection and didn't show neat and clean connections	Few but not all Components are wired with neat and clean connections	Complete components are wired with neat and clean tight connections. Given task completed in due time.
LR3	Troubleshooting	Unable to identify the fault/minimal effort shown	Able to identify the fault but unable to remove it.	Able to identify the fault but partially removes it.	Able to identify the fault and able to make necessary steps and actions to correct it.
LR4	Data Collection	Measurements are incomplete, inaccurate and imprecise. Observations are incomplete or not included. Symbols, units and significant figures are not included	Measurements are somewhat inaccurate and very imprecise. Observations are incomplete or recorded in a confusing way. Major errors using symbols, units and significant digits	Measurements are mostly accurate. Observations are generally complete. Minor errors using symbols, units and significant digits	Measurements are both accurate and precise. Observations are very thorough. Includes appropriate symbols, units and significant digits and completed in due time.
LR5	Results & Plots	Figures, graphs, tables contain errors or are poorly constructed, have missing titles, captions, units missing or incorrect, etc.	Most figures, graphs, tables OK, some still missing some important or required features	All figures, graphs, tables are correctly drawn, but some have minor problems or could still be improved	All figures, graphs, tables are correctly drawn and contain titles/captions.
LR8	Contribution	No participation towards the group tasks	Slight participation towards the group tasks	Substantial participation towards the group tasks	Outstanding participation towards the group tasks
AR4	*Report Submission	Late submission after 1 week.	Late submission after 4 days and within a week.	Late submission after the lab timing and after 2 days of the due date.	Timely submission of the report and in lab time.

***Report:** Report will not be accepted after 1 week of due date